SQL JOINs – Overview

A **JOIN** in SQL is used to combine rows from two or more tables based on a related column between them (usually a foreign key ↔ primary key relationship).

Types of JOINs

1. INNER JOIN

Definition:
Returns only the rows where there is a match in both tables.

Syntax:

```
SELECT columns
FROM table1
INNER JOIN table2
ON table1.common_column = table2.common_column;
```

Behavior:

- Only matching records are returned.
- Non-matching rows are excluded.

Example Use Case:
Get employees who are assigned to a department.

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2. LEFT JOIN (LEFT OUTER JOIN)

Definition:
Returns all rows from the left table and matching rows from the right table. If no match, NULL values are returned for the right table.

Syntax:

```
SELECT columns
FROM table1
LEFT JOIN table2
ON table1.common_column = table2.common_column;
```

Behavior:

- All rows from the left table are included.
- Missing matches from the right table appear as NULL.

Example Use Case:
Get all employees, even those without a department.

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3. RIGHT JOIN (RIGHT OUTER JOIN)

Definition:
Returns all rows from the right table and matching rows from the left table. If no match, NULL values are returned for the left table.

Syntax:

```
SELECT columns
FROM table1
RIGHT JOIN table2
ON table1.common_column = table2.common_column;
```

Behavior:

- All rows from the right table are included.
- Missing matches from the left table appear as NULL.

Example Use Case:
Get all departments, even if no employees are assigned.

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4. FULL JOIN (FULL OUTER JOIN)

Definition:
Returns all rows when there is a match in either left or right table.

Syntax:

```
SELECT columns
FROM table1
FULL OUTER JOIN table2
ON table1.common_column = table2.common_column;
```

Behavior:

- Includes all records from both tables.
- Non-matching rows from either side are filled with NULL.

Example Use Case:
Get all employees and all departments, matched where possible.

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5. CROSS JOIN

Definition:
Returns the Cartesian product of both tables (every row from table1 with every row from table2).

Syntax:

```
SELECT columns
FROM table1
CROSS JOIN table2;
```

Behavior:

- No condition is required.
- Total rows = rows in table1 × rows in table2.

Example Use Case:
Generate all possible combinations (e.g., product variations).

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6. SELF JOIN

Definition:
A table joined with itself.

Syntax:

```
SELECT a.column, b.column
FROM table a
JOIN table b
ON a.common_column = b.common_column;
```

Behavior:

- Uses aliases to differentiate the same table.
- Useful for hierarchical data.

Example Use Case:
Find employees and their managers from the same table.

Quick Comparison Table

JOIN Type	Returns Matching Rows	Includes Non-Matching
INNER JOIN	Yes	No
LEFT JOIN	Yes	Left only
RIGHT JOIN	Yes	Right only
FULL JOIN	Yes	Both sides
CROSS JOIN	Not required	All combinations
SELF JOIN	Depends on condition	Depends

Key Notes

- JOINs rely on a **common column** between tables.
- Always use ON to define the join condition (except CROSS JOIN).
- Improper joins can lead to **duplicate or large datasets**.